

Types of RAG — From Naive to Multimodal Agentic

Type	Core Idea	Example
Naive RAG	Single retrieve → generate pass, flat vector similarity	Lewis et al., 2020
Advanced / Modular RAG	Adds query rewriting, reranking, filtering stages	Improves precision
Graph RAG	Knowledge-graph traversal instead of pure vector search	Microsoft GraphRAG, LightRAG
Agentic RAG	Agent decides when/what/whether to retrieve, can re-plan	Singh et al. survey
Multimodal RAG	Retrieval spans text + image + table, not just text chunks	MuRAG, UniRAG
Multimodal Agentic RAG	Agents plan & reason across modalities (this seminar)	HM-RAG, MultiRAG, ColLEX
Vectorless RAG	Skips embeddings; agent/LLM searches raw text or tools directly	Emerging 2025-26 trend
Cache-Augmented Gen. (CAG)	Preloads knowledge into context/KV-cache ahead of query time	Small, static corpora

15 Papers Reviewed — Summary & Key Findings (1/3)

#	Paper / Venue	Primary Focus	Key Finding
1	HM-RAG <i>ACM Multimedia, 2025</i>	Hierarchical multi-agent multimodal retrieval	Orchestrator + modality-specific sub-agents improve cross-modal retrieval precision over single-agent RAG.
2	MultiRAG <i>Springer LNCS (AI 2025), 2025</i>	Multi-source, multi-modal RAG for scientific research	Dynamic modal weighting preserves figure-text links, boosting accuracy on hybrid queries.
3	Patho-AgenticRAG <i>Preprint, 2025</i>	Domain-specialized agentic RAG for pathology VLMs	RL-based task decomposition & multi-turn search reduce diagnostic hallucination in clinical settings.
4	ColLEX <i>Preprint, 2025</i>	Interactive agentic RAG for scientific collections	User-in-the-loop exploration across images, code, databases, audio & video broadens usable evidence.
5	Agentic RAG (IJCTT) <i>Journal, 2025</i>	AI-driven information retrieval advancement	Frames agentic RAG's advantage over static RAG for dynamic, real-world retrieval tasks.

15 Papers Reviewed — Summary & Key Findings (2/3)

#	Paper / Venue	Primary Focus	Key Finding
6	UniRAGIEEE Access, 2024	Unified retrieval + generation for multimodal QA	Unifying pre-trained LMs for retrieval & generation improves multimodal QA accuracy.
7	Multimodal RAG + KGEmerging Sci. J., 2025	Knowledge-graph-augmented multimodal RAG	Linking multimodal entities via a knowledge graph improves retrieval relevance and cuts noise.
8	Hallucination SurveyACM TOIS (Q1), 2025	Taxonomy of hallucination causes & mitigation in LLMs	Positions retrieval-based grounding as a primary lever for hallucination mitigation.
9	LLM Architectures SurveyIEEE Access, 2024	Benchmarking LLM architecture trends	Identifies efficiency-accuracy trade-offs relevant to choosing an agent's backbone model.
10	Agentic RAG SurveyPreprint (Singh et al.), 2025	Formal taxonomy of agentic RAG	Defines reflection, planning, tool-use & multi-agent collaboration as core agentic RAG patterns.

15 Papers Reviewed — Summary & Key Findings (3/3)

#	Paper / Venue	Primary Focus	Key Finding
11	Ask in Any Modality <i>Preprint, 2025/26</i>	Comprehensive survey of multimodal RAG	Maps modality-specific retrieval techniques and their respective performance gaps.
12	Survey of Multimodal RAG <i>Preprint (Mei et al.), 2025</i>	Broad multimodal RAG survey	Classifies multimodal RAG pipelines by retrieval strategy and fusion method.
13	RAG & Understanding in Vision <i>Preprint, 2025</i>	Vision-centric RAG survey and outlook	Positions vision-grounded retrieval as a needed complement to text-only RAG.
14	Scaling Beyond Context <i>Preprint, 2025</i>	Multimodal RAG for document understanding	Shows RAG-based scaling can outperform long-context-only approaches for document-heavy tasks.
15	mRAG <i>Preprint, 2025</i>	Design space of multimodal RAG	Elucidates architectural choices (retrieval granularity, fusion timing) shaping multimodal RAG performance.